1 (Regression splines, Textbook Ex. 5.4) Consider the truncated power series representation for cubic splines with K-knots $\{\xi_k\}_{k=1}^K \subseteq \mathbb{R}$

$$f(x) = \sum_{j=0}^{3} \beta_j x^j + \sum_{k=1}^{K} \theta_k (x-\xi_k)_+^3. \quad (x \in \mathbb{R})$$

(a) Prove that the natural boundary conditions (*i.e.* f is linear on $(-\infty, \xi_1] \cup [\xi_K, +\infty)$) are equivalent to the coefficient constraints

$$\beta_2 = \beta_3 = \sum_{k=1}^{K} \theta_k = \sum_{k=1}^{K} \xi_k \theta_k = 0.$$

(b) Prove that the truncated power basis under the constraints in (a) can be represented as a linear combination of N_j's

$$N_1(x) = 1, \quad N_2(x) = x, \quad N_{k+2}(x) = d_k(x) - d_{K-1}(x) \quad (x \in \mathbb{R},\ 1 \le k \le K-2)$$

where

$$d_k(x) = \frac{(x-\xi_k)_+^3 - (x-\xi_K)_+^3}{\xi_K - \xi_k} \quad (x \in \mathbb{R},\ 1 \le k \le K-2).$$

Note that $N_j, j = 1, \dots K$ can be used as the set of basis for natural cubic splines.

(a) $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \sum_{k=1}^{K} \theta_k (x-\xi_k)_+^3$

when $x \le \xi_1$: $(x-\xi_k)_+^3 = 0 \Rightarrow f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 \Rightarrow$ linear $\Leftrightarrow \beta_2 = \beta_3 = 0$

when $x \ge \xi_K$: $(x-\xi_k)_+^3 = (x-\xi_k)^3 \Rightarrow f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \sum_{k=1}^{K} \theta_k (x-\xi_k)^3$

$$= \left(\beta_0 - \sum_{k=1}^{K} \theta_k \xi_k^3\right) + \left(\beta_1 + 3\sum_{k=1}^{K} \theta_k \xi_k^2\right) x + \left(\beta_2 - 3\sum_{k=1}^{K} \theta_k \xi_k\right) x^2 + \left(\beta_3 + \sum_{k=1}^{K} \theta_k\right) x^3$$

$\beta_2 = \beta_3 = 0 \Rightarrow$ Linear $\Leftrightarrow \sum_{k=1}^{K} \theta_k \xi_k = \sum_{k=1}^{K} \theta_k = 0$

(b) $f(x) = \beta_0 + \beta_1 x + \sum_{k=1}^{K} \theta_k (x-\xi_k)_+^3$, $\sum_{k=1}^{K} \theta_k \xi_k = \sum_{k=1}^{K} \theta_k = 0$

$$\Rightarrow \theta_{k-1} + \theta_k = -\sum_{k=1}^{k-2} \theta_{k-2}, \quad \xi_{k-1}\theta_{k-1} + \xi_k \theta_k = -\sum_{k=1}^{k-2} \xi_k \theta_k \Rightarrow \begin{cases} \theta_k = -\sum_{k=1}^{k-2} \theta_k - \theta_{k-1} \\ \xi_{k-1}\theta_{k-1} + \xi_k\left(-\sum_{k=1}^{k-2} \theta_k - \theta_{k-1}\right) = -\sum_{k=1}^{k-2} \xi_k \theta_k \end{cases}$$

$$\Rightarrow (\xi_{k-1} - \xi_k)\theta_{k-1} = \sum_{k=1}^{K-2} (\xi_k - \xi_k)\theta_k \Rightarrow \theta_{k-1} = \sum_{k=1}^{K-2} \frac{\xi_k - \xi_k}{\xi_{k-1} - \xi_k} \theta_k \Rightarrow \theta_k = \sum_{k=1}^{K-2} \left(\frac{\xi_k - \xi_k}{\xi_k - \xi_{k-1}} - 1\right)\theta_k = \sum_{k=1}^{K-2} \frac{\xi_{k-1} - \xi_k}{\xi_k - \xi_{k-1}} \theta_k$$

$$g(x) := \sum_{k=1}^{K} \theta_k (x-\xi_k)_+^3 = \sum_{k=1}^{K-2} \theta_k \left[(x-\xi_k)_+^3 - \frac{\xi_k - \xi_k}{\xi_k - \xi_{k-1}}(x-\xi_{k-1})_+^3 + \frac{\xi_{k-1} - \xi_k}{\xi_k - \xi_{k-1}}(x-\xi_k)_+^3\right]$$

For $N_{k+2}(x) = d_k(x) - d_{K-1}(x)$, $d_k(x) = \frac{(x-\xi_k)_+^3 - (x-\xi_K)_+^3}{\xi_K - \xi_k} \Rightarrow N_{k+2}(x) = \frac{(x-\xi_k)_+^3 - (x-\xi_K)_+^3}{\xi_K - \xi_k} - \frac{(x-\xi_{K-1})_+^3 - (x-\xi_K)_+^3}{\xi_K - \xi_{K-1}}$

$$\Rightarrow (\xi_K - \xi_k) N_{k+2}(x) = (x-\xi_k)_+^3 + \left(\frac{\xi_K - \xi_k}{\xi_K - \xi_{K-1}} - 1\right)(x-\xi_K)_+^3 - \left(\frac{\xi_K - \xi_k}{\xi_K - \xi_{K-1}}\right)(x-\xi_{K-1})_+^3$$

$$= (x-\xi_k)_+^3 + \frac{\xi_{K-1} - \xi_k}{\xi_K - \xi_{K-1}}(x-\xi_K)_+^3 - \frac{\xi_K - \xi_k}{\xi_K - \xi_{K-1}}(x-\xi_{K-1})_+^3 \Rightarrow g(x) = \sum_{k=1}^{K-2} \theta_k (\xi_K - \xi_k) N_{k+2}(x)$$

$\Rightarrow f(x) = \beta_0 + \beta_1(x) + g(x) = \beta_0 N_1(x) + \beta_1 N_2(x) + \sum_{k=1}^{K-2} \theta_k (\xi_K - \xi_k) N_{k+2}(x) \Rightarrow N_1, \cdots, N_K$ is the basis of $f(x)$